

HP-optimization → scaling-law deployment workflow

Goal: one consistent, size-matched set of N^* model architectures per dataset size, reusable for ANY reservoir × acquisition combo — find the search recipe ONCE, deploy everywhere.

HP-SEARCH STRATEGY FAMILIES — compare ALL of them

random
(baseline / null)

Optuna-TPE
(Bayesian)

Ray Tune schedulers
(ASHA / BOHB / PBT)

evo_*
(evolutionary, ours)

llm_autoresearch
(LLM AutoResearch)

A · STRATEGY COMPARISON — run every family deep, in isolation

~50 rounds × multiple downsample + HP-init seeds, for every (D × reservoir) cell

TEST GRID
D ∈ {30k, 300k}
×
genomic
motif_planted_v2
dinuc_shuffle

train each config → save val/test preds
+ train_time_sec + LLM \$ / latency → *_meta.json

efficiency curve vs cumulative GPU-SECONDS (fair cost axis)
→ Kneedle + marginal-gain knee, bootstrapped over seeds
→ OPTIMAL # ROUNDS per strategy

rounds knee

GPU-seconds

B · RECIPE COMPOSITION — which strategies work best together

pool ALL harvested configs (best @ each strategy's knee), across strategies

EXHAUSTIVE all-subsets search
ElasticNetCV(positive=True, cv=5)
fit on validation preds
over m = 3, 4, 5, 6, ... strategies

recipe knee

strategies

FROZEN RECIPE
{ winning strategies,
rounds each }
reused at EVERY D

persist every intermediate point (per-round preds, costs, all subset scores) → re-tune knee criteria later

Run the frozen recipe at EACH D → freeze a size-matched architecture set

run FROZEN RECIPE per D
D ∈ {10k, 30k, 100k,
300k, 1M} (+3M later)
over representative reservoirs

forward-selection ElasticNet
curve per D
→ pick ONE global N^* near-knee
ACROSS all D
(size-matched ⇒ fair scaling)

N^* knee

configs (N^*)

- distill to N^* configs / D — select by:
- ORACLE test-Pearson (consistent landscape)
 - HP / architecture diversity
 - reservoir-TYPE coverage

VALIDATE transfer penalty
configs from reservoir A scored on B
vs natively-searched on B
(small gap ⇒ search 1 reservoir/D;
also used as a selection criterion)

DELIVERABLE
per-D table of N^* frozen architectures

DEPLOY: re-train (transfer CONFIGS, not weights)
+ ElasticNet-stack on every
reservoir × acquisition × D — NO HP re-search

All model selection on the ORACLE sequence-function landscape • D=30k prioritized; D=300k preemptible / resumable • one-time per D ⇒ budget can run high